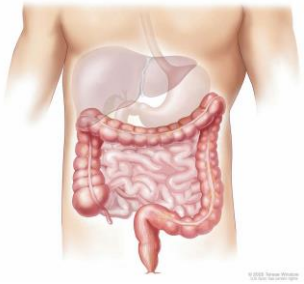
LEMMA
Supply Solutions

BIFIDUMBACTERIUM LONGUM

(Probióticos)



A microflora intestinal é composta por mais de 100 espécies de bactérias benéficas, neutras ou nocivas em permanente atividade. A quantidade e o equilíbrio podem ser alterados por uma grande diversidade de causas, com uso de antibióticos ou de outros medicamentos, o estresse, o envelhecimento ou o consumo de álcool. Neste caso, a saúde do intestino e, consequentemente, a saúde do indivíduo pode ser afetada. Os probióticos como as *bifidobactérium longum* ajudam a restabelecer o equilíbrio da microflora intestinal.

Propriedades & Aplicações:
Estudos demonstram que o *Bifidobacterium longum* melhora a microflora intestinal dos idosos (a quantidade de bifidobactérias diminui com o envelhecimento) que, por seu lado, melhora o ambiente intestinal e contribui para a boa saúde do sistema gastrointestinal e, a sua resistência no combate às infecções. Com a suplementação do probiótico *bidifumbacterium longum*, observa-se:

- ✓ Redução dos produtos de putrefacção, como a amônia;
- ✓ Reduz infecção no lactente prematuro com pouco peso;
- ✓ Estimula a colonização por bifidobactérias e a formação de uma microflora reduzindo assim as infecções intestinais;
- ✓ Leva a um aumento da produção de IgA totais e gera uma resistência à infecção por Escherichia coli;
- ✓ Possui uma ação imunoestimulante (induz a produção de anticorpos específicos e não específicos) e anti-infecciosa (as bifidobactérias produzem ácido acético cuja potente ação bactericida é responsável pela destruição de bactérias nocivas como a Escherichia coli);
- ✓ Previne a diarréia provocada pela administração de eritromicina ou de outras terapias com antibióticos;
- ✓ Regula o trânsito intestinal de indivíduos com constipação reduzindo o teor de amoníaco nas fezes e a atividade de determinadas enzimas fecais e aumentando a frequência das dejecções;
- ✓ Estimula o sistema imunológicos dos idosos imunodeprimidos e ativa neutrófilos e células NK;
- ✓ Reduz o risco de infecções, nomeadamente pelo vírus da gripe;
- ✓ Reduz significativamente os sintomas de alergia ao pólen; e
- ✓ Reforça a resistência óssea, provavelmente reforçando a absorção do cálcio.
- ✓

Posologia Recomendada

A dosagem de Probioicos varia dependendo do produto e indicação específica. Não existe consenso quanto ao número mínimo de microrganismos que devem ser ingeridos para obter um efeito benéfico.

Rua Dr. Arthur Costacurta, 550 – Área Industrial.
Jardinópolis / SP – CEP 14.680-000
Escritório: 11 5044 7234 – São Paulo / SP
www.lemma.com.br – lemma@lemma.com.br

Tipicamente, um probiótico deve conter várias bactérias e milhões de microrganismos para aumentar a probabilidade de colonização intestinal adequada.

Para os lactobacilos, as doses típicas usadas em estudos **variaram 100-2 bilhões UFC/ dia.**

Referencias Bibliográficas

1. Guidelines for the evaluation of probiotics in food: report of a Joint FAO/WHO Working Group. London, Ontario, Canada: Food and Agriculture Organization of the United Nations and World Health Organization; 2002.
2. Senok AC, Ismaeel AY, Botta GA. Probiotics: facts and myths. *Clin Microbiol Infect.* 2005; 11:958–66.
3. Schrezenmeir J, de Vrese M. Probiotics, prebiotics, and synbiotics—approaching a definition. *Am J Clin Nutr*. 2001; 73(suppl):361S-364S.
4. Report of a Joint FAO/WHO Expert Consultation on evaluation of health and nutritional properties of probiotics in food including powder milk with live lactic acid bacteria. Cordoba, Argentina: Food and Agriculture Organization of the United Nations and World Health Organization; 2001.
5. Santosa S, Farnworth E, Jones PJ. Probiotics and their potential health claims. *Nutr Rev*. 2006; 64:265–74.
6. Alvarez-Olmos MI, Oberhelman RA. Probiotic agents and infectious diseases: a modern perspective on a traditional therapy. *Clin Infect Dis*. 2001; 32:1567–76.
7. Doron S, Gorbach SL. Probiotics: their role in the treatment and prevention of disease. *Expert Rev Anti Infect Ther*. 2006; 4:261–75.
8. Pham M, Lemberg DA, Day AS. Probiotics: sorting the evidence from the myths. *Med J Aust*. 2008; 188:304–8.
9. Vanderhoof JA, Young R. Probiotics in the United States. *Clin Infect Dis*. 2008; 46(suppl 2):S67–72.
10. Vanderhoof JA, Young RJ. Current and potential uses of probiotics. *Ann Allergy Asthma Immunol.* 2004; 93(suppl 3):S33–7.
11. Goldin BR. Health benefits of probiotics. *Br J Nutr*. 1998; 80(suppl 2):S203–7.
12. Wald A, Rakel D. Behavioral and complementary approaches for the treatment of irritable bowel syndrome. *Nutr Clin Pract*. 2008; 23:284–92.
13. Surawicz CM. Role of probiotics in antibiotic-associated diarrhea, *Clostridium difficile*-associated diarrhea, and recurrent *Clostridium difficile*-associated diarrhea. *J Clin Gastroenterol*. 2008; 42(suppl 2):S64–70.
14. MacIntyre A, Cymet TC. Probiotics: the benefits of bacterial cultures. *Compr Ther*. 2005; 31:181–5.

15. Scarpellini E, Cazzato A, Lauritano C et al. Probiotics: which and when? *Dig Dis*. 2008; 26:175–82.

16. Farnworth ER. The evidence to support health claims for probiotics. *J Nutr*. 2008; 138(suppl):1250S–4S.

17. Goldin BR, Gorbach SL. Clinical indications for probiotics: an overview. *Clin Infect Dis*. 2008; 46(suppl 2):S96–100.

18. Reid G, Jass J, Sebulsky MT et al. Potential uses of probiotics in clinical practice. *Clin Microbiol Rev*. 2003; 16:658–72.

19. Silva M, Jacobus NV, Deneke C et al. Antimicrobial substance from a human *Lactobacillus* strain. *Antimicrob Agents Chemother*. 1987; 31:1231–3.

20. Castagliuolo I, Riegler MF, Valenick L et al. *Saccharomyces boulardii* protease inhibits the effects of *Clostridium difficile* toxins A and B in human colonic mucosa. *Infect Immun*. 1999; 67:302–7.

21. Chan RC, Reid G, Irvin RT et al. Competitive exclusion of uropathogens from human uroepithelial cells by *Lactobacillus* whole cells and cell wall fragments. *Infect Immun*. 1985; 47:84–9.

22. Mack DR, Michail S, Wei S et al. Probiotics inhibit enteropathogenic *E. coli* adherence in vitro by inducing intestinal mucin gene expression. *Am J Physiol*. 1999; 276:G941–50.

23. Kaila M, Isolauri E, Soppi E et al. Enhancement of the circulating antibody secreting cell response in human diarrhea by a human *Lactobacillus* strain. *Pediatr Res*. 1992; 32:141–4.

24. Probiotics. *Med Lett Drugs Ther*. 2007; 49:66–8.

25. Guandalini S, Pensabene L, Zikri MA et al. *Lactobacillus* GG administered in oral rehydration solution to children with acute diarrhea: a multicenter European trial. *J Pediatr Gastroenterol Nutr*. 2000; 30:54–60.

26. Van Niel CW, Feudtner C, Garrison MM et al. *Lactobacillus* therapy for acute infectious diarrhea in children: a metaanalysis. *Pediatrics*. 2002; 109:678–84.

27. Canani RB, Cirillo P, Terrin G et al. Probiotics for treatment of acute diarrhoea in children: randomised clinical trial of five different preparations. *BMJ*. 2007; 335:340–2.

28. Cremonini F, Di Caro S, Nista EC et al. Meta-analysis: the effect of probiotic administration on antibiotic-associated diarrhoea. *Aliment Pharmacol Ther*. 2002; 16:1461–7.

29. D'Souza AL, Rajkumar C, Cooke J et al. Probiotics in prevention of antibiotic associated diarrhoea: meta-analysis. *BMJ*. 2002; 324:1361–6.

30. McFarland LV. Meta-analysis of probiotics for the prevention of antibiotic associated diarrhea and the treatment of *Clostridium difficile* disease. *Am J Gastroenterol*. 2006; 101:812–22.

31. Klarin B, Wullt M, Palmquist I et al. *Lactobacillus plantarum* 299v reduces colonisation of *Clostridium difficile* in critically ill patients treated with antibiotics. *Acta Anaesthesiol Scand*. 2008; 52:1096–102.

32. Dendukuri N, Costa V, McGregor M et al. Probiotic therapy for the prevention and treatment of *Clostridium difficile*associated diarrhea: a systematic review. *CMAJ*. 2005; 173:167–70.

33. Pillai A, Nelson RL. Probiotics for treatment of *Clostridium difficile*-associated colitis in adults. *Cochrane Database Syst Rev*. 2008; 1:CD004611.

34. Hilton E, Kolakowski P, Singer C et al. Efficacy of *Lactobacillus* GG as a diarrheal preventive in travelers. *J Travel Med*. 1997; 4:41–3.

35. Katelaris PH, Salam I, Farthing MJ. Lactobacilli to prevent traveler's diarrhea? *N Engl J Med*. 1995; 333:1360–1. Letter.

36. McFarland LV. Meta-analysis of probiotics for the prevention of traveler's diarrhea. *Travel Med Infect Dis*. 2007; 5:97–105.

37. Wilhelm SM, Brubaker CM, Varcak EA et al. Effectiveness of probiotics in the treatment of irritable bowel syndrome. *Pharmacotherapy*. 2008; 28:496–505.

38. McFarland LV, Dublin S. Meta-analysis of probiotics for the treatment of irritable bowel syndrome. *World J Gastroenterol*. 2008; 14:2650–61.

39. Rembacken BJ, Snelling AM, Hawkey PM et al. Non-pathogenic *Escherichia coli* versus mesalazine for the treatment of ulcerative colitis: a randomised trial. *Lancet*. 1999; 354:635–9.

40. Kruis W, Schutz E, Fric P et al. Double-blind comparison of an oral *Escherichia coli* preparation and mesalazine in maintaining remission of ulcerative colitis. *Aliment Pharmacol Ther*. 1997; 11:853–8.

41. Kruis W, Fric P, Pokrotnieks J et al. Maintaining remission of ulcerative colitis with the probiotic *Escherichia coli* Nissle 1917 is as effective as with standard mesalazine. *Gut*. 2004; 53:1617–23.

42. Guslandi M, Giollo P, Testoni PA. A pilot trial of *Saccharomyces boulardii* in ulcerative colitis. *Eur J Gastroenterol Hepatol*. 2003; 15:697–8.

43. Bibiloni R, Fedorak RN, Tannock GW et al. VSL#3 probiotic-mixture induces remission in patients with active ulcerative colitis. *Am J Gastroenterol*. 2005; 100:1539–46.

44. Guslandi M, Mezzi G, Sorghi M et al. *Saccharomyces boulardii* in maintenance treatment of Crohn's disease. *Dig Dis Sci*. 2000; 45:1462–4.

45. Marteau P, Lemann M, Seksik P et al. Ineffectiveness of *Lactobacillus johnsonii* LA1 for prophylaxis of postoperative recurrence in Crohn's disease: a randomised, double blind, placebo controlled GETAID trial. *Gut*. 2006; 55:842–7.

46. Gionchetti P, Rizzello F, Venturi A et al. Oral bacteriotherapy as maintenance treatment in patients with chronic pouchitis: a double-blind, placebo-controlled trial. *Gastroenterology*. 2000; 119:305–9.

47. Gionchetti P, Rizzello F, Helwig U et al. Prophylaxis of pouchitis onset with probiotic therapy: a double-blind, placebo-controlled trial. *Gastroenterology*. 2003; 124:1202–9.

48. Mimura T, Rizzello F, Helwig U et al. Once daily high dose probiotic therapy (VSL#3) for maintaining remission in recurrent or refractory pouchitis. *Gut*. 2004; 53:108–14.

49. Kuisma J, Mentula S, Jarvinen H et al. Effect of *Lactobacillus rhamnosus* GG on ileal pouch inflammation and microbial flora. *Aliment Pharmacol Ther*. 2003; 17:509–15.

50. Kalliomaki M, Salminen S, Arvilommi H et al. Probiotics in primary prevention of atopic disease: a randomised placebo-controlled trial. *Lancet*. 2001; 357:1076–9.

51. Kalliomaki M, Salminen S, Poussa T et al. Probiotics and prevention of atopic disease: 4-year follow-up of a randomised placebo-controlled trial. *Lancet*. 2003; 361:1869–71.

52. Isolauri E, Arvola T, Sutas Y et al. Probiotics in the management of atopic eczema. *Clin Exp Allergy*. 2000; 30:1604–10.

53. Reid G, Bruce AW. Urogenital infections in women: can probiotics help? *Postgrad Med J*. 2003; 79:428–32.

54. Hilton E, Rindos P, Isenberg HD. *Lactobacillus* GG vaginal suppositories and vaginitis. *J Clin Microbiol*. 1995; 33:1433. Letter.

55. Reid G, Bruce AW, Fraser N et al. Oral probiotics can resolve urogenital infections. *FEMS Immunol Med Microbiol*. 2001; 30:49–52.

56. Hilton E, Isenberg HD, Alperstein P et al. Ingestion of yogurt containing *Lactobacillus acidophilus* as prophylaxis for candidal vaginitis. *Ann Intern Med*. 1992; 116:353–7.

57. Kligler B, Cohrssen A. Probiotics. *Am Fam Physician*. 2008; 78:1073–8.

58. Natural Medicines Comprehensive Database. *Lactobacillus* monograph. www.naturaldatabase.com (accessed 2009 Mar 23).

59. Sutton A. Product development of probiotics as biological drugs. *Clin Infect Dis*. 2008; 46(suppl 2):S128–32.

60. Kailasapathy K. Microencapsulation of probiotic bacteria: technology and potential applications. *Curr Issues Intest Microbiol*. 2002; 3:39–48.

61. Katz JA, Pirovano F, Matteuzzi D et al. Commercially available probiotic preparations: are you getting what you pay for? *Gastroenterology*. 2002; 122(suppl 1): A-459. Abstract.

62. Sanders ME. Probiotics: background and product table. www.usprobiotics.org (accessed 2009 Mar 23).

63. Natural Medicines Comprehensive Database. Yogurt monograph. www.naturaldatabase.com (accessed 2009 Mar 23).

64. Karpa KD. Probiotics for *Clostridium difficile* diarrhea: putting it into perspective. *Ann Pharmacother*. 2007; 41:1284–7.

65. Snydman DR. The safety of probiotics. *Clin Infect Dis*. 2008; 46(suppl 2):S104–11.

66. Borriello SP, Hammes WP, Holzapfel W et al. Safety of probiotics that contain lactobacilli or bifidobacteria. *Clin Infect Dis*. 2003; 36:775–80.

67. Salminen MK, Rautelin H, Tynkkynen S et al. *Lactobacillus* bacteremia, clinical significance, and patient outcome, with special focus on probiotic *L. rhamnosus* GG. *Clin Infect Dis*. 2004; 38:62–9.

68. Munoz P, Bouza E, Cuenca-Estrella M et al. *Saccharomyces cerevisiae* fungemia: an emerging infectious disease. *Clin Infect Dis*. 2005; 40:1625–34.

69. Boyle RJ, Robins-Browne RM, Tang ML. Probiotic use in clinical practice: what are the risks? *Am J Clin Nutr*. 2006; 83:1256–64.

70. Salminen MK, Tynkkynen S, Rautelin H et al. *Lactobacillus* bacteremia during a rapid increase in probiotic use of *Lactobacillus rhamnosus* GG in Finland. *Clin Infect Dis*. 2002; 35:1155–60.

71. Besselink MG, van Santvoort HC, Buskens E et al. Probiotic prophylaxis in predicted severe acute pancreatitis: a randomised, double-blind, placebo-controlled trial. *Lancet*. 2008; 371:651–9.

72. Natural Medicines Comprehensive Database. Bifidobacteria monograph. www.naturaldatabase.com (accessed 2009 Mar 23).

73. Natural Medicines Comprehensive Database. *Saccharomyces boulardii* monograph. www.naturaldatabase.com (accessed 2009 Mar 23).

74. Florastor (*Saccharomyces boulardii* lyo) package insert. Beauvais, France: Biocodex; 2005–09.

75. Drugdex System. *Lactobacillus* monograph. Greenwood Village, CO: Thomson Micromedex (accessed 2009 Mar 23).

76. Drugdex System. *Saccharomyces boulardii* monograph. Greenwood Village, CO: Thomson Micromedex (accessed 2009 Mar 23).

77. BENJAMIN KLIGLER, MD, MPH, is associate professor of family medicine at the Albert Einstein College of Medicine of Yeshiva University in New York, NY. He received his medical degree from Boston (Mass.) University School of Medicine and completed a residency in family medicine at Montefiore Medical Center in New York, NY.

78. ANDREAS COHRSSEN, MD, is program director of the Beth Israel Residency Program in Urban Family Practice in New York, NY. He received his medical degree from the Johann Wolfgang Goethe—Universität Frankfurt (Germany), and completed a residency in family medicine at Lutheran Medical Center in New York, NY.

79. *Address correspondence to Benjamin Kligler, MD, MPH, Continuum Center for Health and Healing, 245 Fifth Ave., 2nd Floor, New York, NY 10016 (e-mail: bkligler@chpnet.org). Reprints are not available from the authors.*

80. Food and Agriculture Organization of the United Nations and the World Health Organization. Health and nutritional properties of probiotics in food including powder milk with live lactic acid bacteria. Cordoba, Argentina; 2001.

http://www.who.int/foodsafety/publications/fs_management/en/probiotics.pdf. Accessed July 2, 2008.

81. 2. Prescott SL, Dunstan JA, Hale J, et al. Clinical effects of probiotics are associated with increased interferon-gamma responses in very young children with atopic dermatitis. *Clin Exp Allergy*. 2005;35(12):1557–1564.

82. 3. Doron S, Snydman DR, Gorbach SL. *Lactobacillus* GG: bacteriology and clinical applications. *Gastroenterol Clin North Am*. 2005;34(3):483–498.

83. 4. Rosenfeldt V, Benfeldt E, Valerius NH, Paerregaard A, Michaelsen KF. Effect of probiotics on gastrointestinal symptoms and small intestinal permeability in children with atopic dermatitis. *J Pediatr*. 2004;145(5):612–616.

84. 5. Sazawal S, Hiremath G, Dhingra U, Malik P, Deb S, Black RE. Efficacy of probiotics in prevention of acute diarrhoea: a meta-analysis of masked, randomised, placebo-controlled trials. *Lancet Infect Dis*. 2006;6(6):374–382.

85. 6. McFarland LV. Meta-analysis of probiotics for the prevention of antibiotic associated diarrhea and the treatment of *Clostridium difficile* disease. *Am J Gastroenterol*. 2006;101(4):812–822.

86. 7. Szajewska H, Ruszczyński M, Radzikowski A. Probiotics in the prevention of antibiotic-associated diarrhea in children: a meta-analysis of randomized controlled trials. *J Pediatr*. 2006;149(3):367–372.

87. 8. Johnston BC, Supina AL, Ospina M, Vohra S. Probiotics for the prevention of pediatric antibiotic-associated diarrhea. *Cochrane Database Syst Rev*. 2007;(2):CD004827.

88. 9. Hickson M, D'Souza AL, Muthu N, et al. Use of probiotic *Lactobacillus* preparation to prevent diarrhoea associated with antibiotics: randomised double blind placebo controlled trial. *BMJ*. 2007;335(7610):80.

89. 10. Allen SJ, Okoko B, Martinez E, Gregorio G, Dans LF. Probiotics for treating infectious diarrhoea. *Cochrane Database of Syst Rev*. 2004;(2)CD003048.

90. 11. Szajewska H, Skórka A, Dylag M. Meta-analysis: *Saccharomyces boulardii* for treating acute diarrhoea in children. *Aliment Pharmacol Ther*. 2007;25(3):257–264.

91. 12. Canani RB, Cirillo P, Terrin G, et al. Probiotics for treatment of acute diarrhoea in children: randomised clinical trial of five different preparations. *BMJ*. 2007;335(7615):340.

92. 13. Weizman Z, Asli G, Alsheikh A. Effect of a probiotic formula on infections in child care centers: comparison of two probiotic agents. *Pediatrics*. 2005;115(1):5–9.

93. 14. Pedone CA, Arnaud CC, Postaire ER, Bouley CF, Reinert P. Multicentric study of the effect of milk fermented by *Lactobacillus casei* on the incidence of diarrhoea. *Int J Clin Pract*. 2000;54(9):568–571.

94. 15. Pedone CA, Bernabeu AO, Postaire ER, Bouley CF, Reinert P. The effect of supplementation with milk fermented by *Lactobacillus casei* (strain DN-114 001) on acute diarrhoea in children attending day care centres. *Int J Clin Pract*. 1999;53(3):179–184.

95. 16. McFarland LV. Meta-analysis of probiotics for the prevention of traveler's diarrhea. *Travel Med Infect Dis*. 2007;5(2):97–105.

96. 17. Whorwell PJ, Altringer L, Morel J, et al. Efficacy of an encapsulated probiotic *Bifidobacterium infantis* 35624 in women with irritable bowel syndrome. *Am J Gastroenterol*. 2006;101(7):1581–1590.

97. 18. Bausserman M, Michail S. The use of *Lactobacillus* GG in irritable bowel syndrome in children: a double-blind randomized control trial. *J Pediatr*. 2005;147(2):197–201.

98. . Guyonnet D, Chassany O, Ducrotte P, et al. Effect of a fermented milk containing *Bifidobacterium animalis* DN-173 010 on the health-related quality of life and symptoms in irritable bowel syndrome in adults in primary care: a multicentre, randomized, double-blind, controlled trial. *Aliment Pharmacol Ther*. 2007;26(3):475–486.

99. Kalliomäki M, Salminen S, Arvilommi H, Kero P, Koskinen P, Isolauri E. Probiotics in primary prevention of atopic disease: a randomised placebo-controlled trial. *Lancet*. 2001;357(9262):1076–1079.

100. Kalliomäki M, Salminen S, Poussa T, Arvilommi H, Isolauri E. Probiotics and prevention of atopic disease: 4-year follow-up of a randomised placebo-controlled trial. *Lancet*. 2003;361(9372):1869–1871.

101. Kalliomäki M, Salminen S, Poussa T, Isolauri E. Probiotics during the first 7 years of life: a cumulative risk reduction of eczema in a randomized, placebo-controlled trial. *J Allergy Clin Immunol*. 2007;119(4):1019–1021.

102. Kukkonen K, Savilahti E, Haahtela T, et al. Probiotics and prebiotic galacto-oligosaccharides in the prevention of allergic diseases: a randomized, double-blind, placebo-controlled trial. *J Allergy Clin Immunol*. 2007;119(1):192–198.

103. Taylor AL, Dunstan JA, Prescott SL. Probiotic supplementation for the first 6 months of life fails to reduce the risk of atopic dermatitis and increases the risk of allergen sensitization in high-risk children: a randomized controlled trial. *J Allergy Clin Immunol*. 2007;119(1):184–191.

104. Viljanen M, Savilahti E, Haahtela T, et al. Probiotics in the treatment of atopic eczema/dermatitis syndrome in infants: a double-blind placebo-controlled trial. *Allergy*. 2005;60(4):494–500.

105. Sistek D, Kelly R, Wickens K, Stanley T, Fitzharris P, Crane J. Is the effect of probiotics on atopic dermatitis confined to food sensitized children? *Clin Exp Allergy*. 2006;36(5):629–633.

106. Passeron T, Lacour JP, Fontas E, Ortonne JP. Prebiotics and synbiotics: two promising approaches for the treatment of atopic dermatitis in children above 2 years. *Allergy*. 2006;61(4):431–437.

107. Weston S, Halbert A, Richmond P, Prescott SL. Effects of probiotics on atopic dermatitis: a randomised controlled trial. *Arch Dis Child*. 2005;90(9):892–897.

108. Hammerman C, Bin-Nun A. Safety of probiotics: comparison of two popular strains. *BMJ*. 2006;333(7576):1006–1008.

ConsumerLab.com product review: probiotic supplements (including *Lactobacillus acidophilus*, *Bifidobacterium*, and others).
http://www.consumerlab.com/results/probiotics.asp